

11. Sequences and Series

11.1. Sequences

11.1 Learning Objectives

- Understand how to write a sequence and find their formula if they have one
- Find the limit of a sequence using limit laws and theorems
- Determine whether sequences converge or diverge using theorems
- Determine whether a sequence is increasing or decreasing, and whether it is bounded

Definition 11.1.1 Sequence

A **sequence** is a list of numbers $a_1, a_2, a_3, \dots, a_n, \dots$ in a definite order.

The a_i are called the **terms (or values)** of the sequence.

Consider the sequence

$$1, 1, 2, 3, 5, 8, 13, \dots$$

What is a_1 ?

$$\underbrace{1}_{a_1}, 1, 2, 3, 5, 8, 13, \dots$$

This is the first term in the list, so it is $a_1 = 1$.

What is a_6 ?

$$1, 1, 2, 3, 5, \underbrace{8}_{a_6}, 13, \dots$$

This is the sixth term in the list, so it is $a_6 = 8$.

Definition 11.1.2

The integer n is called the **index** of a_n . It indicates what term in the sequence you are considering.

We can think of the sequence $a_1, a_2, \dots, a_n, \dots$ as a function that sends the positive integer n to the n^{th} value in the sequence, a_n .

$$\begin{aligned} f(n) &= a_n, \text{ so } f(1) = a_1 \\ f(2) &= a_2 \\ f(3) &= a_3 \\ &\vdots \end{aligned}$$

The length of a sequence refers to the **number of terms** in the sequence. Sequences can be either infinite or finite.

Note

Notation: We typically denote sequences with some kind of parentheses or brackets.

$$(a_n)_{n=1}^{\infty}, [a_n]_{n=1}^{\infty}, \{a_n\}_{n=1}^{\infty}$$

all refer to the sequence

$$(a_1, a_2, a_3, \dots, a_n, \dots)$$

where a_i is the i^{th} term in the sequence.

Note: the indexing does not have to start at $n = 1$, and sometimes the index labeling is left off (something like just “ (a_n) ” for example).

Example 11.1.1

List the first 6 terms of the following sequences

1. $(a_n)_{n=1}^{\infty} = (\sqrt{n})_{n=1}^{\infty}$
2. $(b_n)_{n=3}^{\infty} = (\cos(\pi n))_{n=3}^{\infty}$

Solution

1. $(a_n)_{n=1}^{\infty} = (\sqrt{n})_{n=1}^{\infty}$

$$a_1 = 1, a_2 = \sqrt{2}, a_3 = \sqrt{3}, a_4 = 2, a_5 = \sqrt{5}, a_6 = \sqrt{6}$$

1. $(b_n)_{n=3}^{\infty} = (\cos(\pi n))_{n=3}^{\infty}$

$$b_3 = \cos(3\pi) = -1, b_4 = \cos(4\pi) = 1, b_5 = -1, b_6 = 1, b_7 = -1, b_8 = 1$$

Just because the indexing starts at $n = 3$, doesn't change how we read the first 6 terms.

Example 11.1.2

Write a rule that specifies the values of the sequence

$$(b_n)_{n=1}^{\infty} = \left(-\frac{1}{2}, \frac{2}{3}, -\frac{3}{4}, \frac{4}{5}, \dots\right).$$

Solution

Notice, the numerator is the index, and the denominator is one larger than the index. So, accounting for the alternating $\pm$ signs,

$$b_n = (-1)^n \frac{n}{n+1}.$$

Definition 11.1.3

A **recurrence relation** for a sequence $a_0, a_1, a_2, \dots$ is a formula that relates each value a_k to certain predecessors $a_{k-1}, a_{k-2}, \dots$.

Example 11.1.3

Define a sequence $c_0, c_1, c_2, \dots$ recursively as follows: For all integers $k \geq 2$,

1. $c_k = c_{k-1} + k c_{k-2} + 1$, we call this the recurrence relation
2. $c_0 = 1$ and $c_1 = 2$, we call this the initial conditions

Find c_2, c_3, c_4 .

Solution

$$c_2 = c_1 + 2 \cdot c_0 + 1 = 2 + 2(1) + 1 = 5$$

$$c_3 = c_2 + 3 \cdot c_1 + 1 = 5 + 3(2) + 1 = 12$$

$$c_4 = c_3 + 4 \cdot c_2 + 1 = 12 + 4(5) + 1 = 33$$

Definition 11.1.4

The sequence (a_n) **converges** to the number L if $\lim_{n \rightarrow \infty} a_n = L$.

If no such number L exists, we say that (a_n) **diverges**.

We say the sequence (a_n) **diverges to (minus) infinity** if $\lim_{n \rightarrow \infty} a_n = \pm\infty$

What does this mean?

If we go back to thinking about a sequence as a function that maps integers to real numbers, then we can essentially think of the terms in the sequence as being part of a continuous function with the same behavior as the sequence at integers. Then we can look at the limit of the function.

Theorem 11.1.1

If $\lim_{x \rightarrow \infty} f(x) = L$ and $f(n) = a_n$ when n is an integer, then

$$\lim_{n \rightarrow \infty} a_n = L.$$

Example 11.1.4

Does the sequence $\left(\frac{n^2 + 3n + 1}{2n^2 + 1} \right)_{n=1}^{\infty}$ converge?

Solution

It suffices to check that $\lim_{x \rightarrow \infty} \frac{x^2 + 3x + 1}{2x^2 + 1}$ converges, so we can look at this and use regular limit techniques.

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{x^2 + 3x + 1}{2x^2 + 1} & \text{ has IFL } \frac{\infty}{\infty} \\
 & \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{2x + 3}{4x} \text{ has IFL } \frac{\infty}{\infty} \\
 & \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{2}{4} \\
 & = \frac{1}{2}
 \end{aligned}$$

Because the function where $f(n) = a_n$ converges to $\frac{1}{2}$, the sequence $(a_n)_{n=1}^{\infty} = \left(\frac{n^2 + 3n + 1}{2n^2 + 1} \right)_{n=1}^{\infty}$ converges $\frac{1}{2}$ as well.

11.1a Section Summary: